THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 100°	Find x first.
2	**(d)** 110°	Find B, then A.
3	**(b)** 90°	Half of 180°.
4	**(c)** 45°	Equate.
5	**(d)** 100°	$x + y = 180$, $x - y = d$.
6	**(a)** 60°	Share 90° by the ratio.
7	**(d)** perpendicular to the other	Reason with angle relations.
8	**(b)** 95°	Find x first.
9	**(c)** 125°	Find B, then A.
10	**(c)** parallel	Use linear pairs and parallel-line properties.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a + c = 180^\circ$ and $b + c = 180^\circ$; Hence $a = b$ (1); With $c = 40^\circ$: $a = b = 140^\circ$ (1).	1+1
12	Corresponding angles are equal; $5x - 10 = 4x + 15$, so $x = 25$ (1); $\angle 1 = 115^\circ$ (1).	1+1
13	$180 - x = 3(90 - x)$; $(3 - 1)x = 90$ (1); $x = 45^\circ$ (1).	1+1
14	$a + c = 180^\circ$ and $b + c = 180^\circ$; Hence $a = b$ (1); With $c = 75^\circ$: $a = b = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x + 10 = x + 40$ gives $x = 30$ (1); Angle $= 70^\circ$ (1); Supplementary partner $= 110^\circ$ (1).	1+1+1
16	Angles on a line sum to 180°; $4x + 28 = 180$ (1); $x = 38$ (1); Angles 38°, 48°, 74°, 20° (1).	1+1+1
17	Alternate interior angles equal, so halves equal; Halves 20° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $20^\circ = 20^\circ$ (1).	1+1+1
	OR: $2x + 15 = x + 55$ gives $x = 40$ (1); Angle $= 95^\circ$ (1); Supplementary partner $= 85^\circ$ (1).	1+1+1
18	Angles on a line sum to 180°; $4x + 32 = 180$ (1); $x = 37$ (1); Angles 37°, 49°, 70°, 24° (1).	1+1+1
	OR: Alternate interior angles equal, so halves equal; Halves 25° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $25^\circ = 25^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Co-interior angles sum to 180° (1); $9x + 9 = 180$, so $x = 19$ (1); $\angle 1 = 86^\circ$ (1); $\angle 2 = 94^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 3 (1); Angles 60° and 120° (1); Corresponding angle of larger $= 120^\circ$ (1); Vertically opposite of smaller $= 60^\circ$ (1); Reasons stated (1).	1+1+1+1+1
20	Co-interior angles sum to 180° (1); $9x + 18 = 180$, so $x = 18$ (1); $\angle 1 = 92^\circ$ (1); $\angle 2 = 88^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 5 (1); Angles 72° and 108° (1); Corresponding angle of larger $= 108^\circ$ (1); Vertically opposite of smaller $= 72^\circ$ (1); Reasons stated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 75°	Equate.
2	**(b)** 110°	$x + y = 180$, $x - y = d$.
3	**(a)** 54°	Share 90° by the ratio.
4	**(a)** parallel	Reason with angle relations.
5	**(c)** 108°	Find x first.
6	**(d)** 140°	Find B, then A.
7	**(b)** 90°	Use linear pairs and parallel-line properties.
8	**(b)** 105°	Equate.
9	**(b)** 120°	$x + y = 180$, $x - y = d$.
10	**(c)** 50°	Share 90° by the ratio.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Corresponding angles are equal; $5x - 20 = 4x + 5$, so $x = 25$ (1); $\angle 1 = 105^\circ$ (1).	1+1
12	$180 - x = 4(90 - x)$; $(4 - 1)x = 180$ (1); $x = 60^\circ$ (1).	1+1
13	$a + c = 180^\circ$ and $b + c = 180^\circ$; Hence $a = b$ (1); With $c = 110^\circ$: $a = b = 70^\circ$ (1).	1+1
14	Corresponding angles are equal; $5x - 8 = 4x + 12$, so $x = 20$ (1); $\angle 1 = 92^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x + 20 = x + 60$ gives $x = 40$ (1); Angle $= 100^\circ$ (1); Supplementary partner $= 80^\circ$ (1).	1+1+1
16	Angles on a line sum to 180°; $4x + 32 = 180$ (1); $x = 37$ (1); Angles 37°, 57°, 70°, 16° (1).	1+1+1
17	Alternate interior angles equal, so halves equal; Halves 30° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $30^\circ = 30^\circ$ (1).	1+1+1
	OR: $2x + 5 = x + 35$ gives $x = 30$ (1); Angle $= 65^\circ$ (1); Supplementary partner $= 115^\circ$ (1).	1+1+1
18	Angles on a line sum to 180°; $4x + 36 = 180$ (1); $x = 36$ (1); Angles 36°, 44°, 70°, 30° (1).	1+1+1
	OR: Alternate interior angles equal, so halves equal; Halves 35° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $35^\circ = 35^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Co-interior angles sum to 180° (1); $9x + 27 = 180$, so $x = 17$ (1); $\angle 1 = 98^\circ$ (1); $\angle 2 = 82^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 6 (1); Angles 30° and 150° (1); Corresponding angle of larger $= 150^\circ$ (1); Vertically opposite of smaller $= 30^\circ$ (1); Reasons stated (1).	1+1+1+1+1
20	Co-interior angles sum to 180° (1); $9x + 36 = 180$, so $x = 16$ (1); $\angle 1 = 104^\circ$ (1); $\angle 2 = 76^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 9 (1); Angles 80° and 100° (1); Corresponding angle of larger $= 100^\circ$ (1); Vertically opposite of smaller $= 80^\circ$ (1); Reasons stated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $l \parallel n$	Reason with angle relations.
2	**(a)** 98°	Find x first.
3	**(c)** 155°	Find B, then A.
4	**(b)** 45°	Use linear pairs and parallel-line properties.
5	**(c)** 135°	Equate.
6	**(c)** 140°	$x + y = 180$, $x - y = d$.
7	**(c)** 75°	Share 90° by the ratio.
8	**(a)** 180°	Reason with angle relations.
9	**(d)** equal	Use linear pairs and parallel-line properties.
10	**(b)** equal	Use linear pairs and parallel-line properties.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$180 - x = 6(90 - x)$; $(6 - 1)x = 360$ (1); $x = 72^\circ$ (1).	1+1
12	$a + c = 180^\circ$ and $b + c = 180^\circ$; Hence $a = b$ (1); With $c = 140^\circ$: $a = b = 40^\circ$ (1).	1+1
13	Corresponding angles are equal; $5x - 25 = 4x + 10$, so $x = 35$ (1); $\angle 1 = 150^\circ$ (1).	1+1
14	$180 - x = 7(90 - x)$; $(7 - 1)x = 450$ (1); $x = 75^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x + 25 = x + 65$ gives $x = 40$ (1); Angle $= 105^\circ$ (1); Supplementary partner $= 75^\circ$ (1).	1+1+1
16	Angles on a line sum to 180°; $4x + 24 = 180$ (1); $x = 39$ (1); Angles 39°, 45°, 76°, 20° (1).	1+1+1
17	Alternate interior angles equal, so halves equal; Halves 40° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $40^\circ = 40^\circ$ (1).	1+1+1
	OR: $2x + 12 = x + 48$ gives $x = 36$ (1); Angle $= 84^\circ$ (1); Supplementary partner $= 96^\circ$ (1).	1+1+1
18	Angles on a line sum to 180°; $4x + 40 = 180$ (1); $x = 35$ (1); Angles 35°, 49°, 68°, 28° (1).	1+1+1
	OR: Alternate interior angles equal, so halves equal; Halves 45° are alternate angles on the bisectors (1); Equal alternate angles imply parallel (1); Check: $45^\circ = 45^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Co-interior angles sum to 180° (1); $9x + 18 = 180$, so $x = 18$ (1); $\angle 1 = 90^\circ$ (1); $\angle 2 = 90^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 4 (1); Angles 45° and 135° (1); Corresponding angle of larger $= 135^\circ$ (1); Vertically opposite of smaller $= 45^\circ$ (1); Reasons stated (1).	1+1+1+1+1
20	Co-interior angles sum to 180° (1); $9x + 18 = 180$, so $x = 18$ (1); $\angle 1 = 97^\circ$ (1); $\angle 2 = 83^\circ$ and $\angle 3 = \angle 1$ (1); Reasons stated (1).	1+1+1+1+1
	OR: Sum 180°, parts 9 (1); Angles 40° and 140° (1); Corresponding angle of larger $= 140^\circ$ (1); Vertically opposite of smaller $= 40^\circ$ (1); Reasons stated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.